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Tutorial Video: Defects

Differences among Point, Linear, and Interfacial Defects

of material between these boundaries is appropriately termed a *twin*. Twins result from atomic displacements that are produced from applied mechanical shear forces (mechanical twins) and also during annealing heat treatments following deformation (annealing twins). Twinning occurs on a definite crystallographic plane and in a specific direction, both of which depend on the crystal structure. Annealing twins are typically found in metals that have the FCC crystal structure, whereas mechanical twins are observed in BCC and HCP metals. The role of mechanical twins in the deformation process is discussed in Section 7.7. Annealing twins may be observed in the photomicrograph of the polycrystalline brass specimen shown in Figure 4.14*c*. The twins correspond to those regions having relatively straight and parallel sides and a different visual contrast than the untwinned regions of the grains within which they reside. An explanation for the variety of textural contrasts in this photomicrograph is provided in Section 4.10.

Miscellaneous Interfacial Defects

Other possible interfacial defects include stacking faults and ferromagnetic domain walls. Stacking faults are found in FCC metals when there is an interruption in the *ABCABCABC* . . . stacking sequence of close-packed planes (Section 3.12). For ferromagnetic and ferrimagnetic materials, the boundary that separates regions having different directions of magnetization is termed a *domain wall,* which is discussed in Section 20.7.

Associated with each of the defects discussed in this section is an interfacial energy, the magnitude of which depends on boundary type, and which varies from material to material. Normally, the interfacial energy is greatest for external surfaces and least for domain walls.

Concept Check 4.3 The surface energy of a single crystal depends on crystallographic orientation. Does this surface energy increase or decrease with an increase in planar density? Why?

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister* (*Student Companion Site*).]

4.7 BULK OR VOLUME DEFECTS

Other defects exist in all solid materials that are much larger than those heretofore discussed. These include pores, cracks, foreign inclusions, and other phases. They are normally introduced during processing and fabrication steps. Some of these defects and their effects on the properties of materials are discussed in subsequent chapters.

4.8 ATOMIC VIBRATIONS

atomic vibration

Every atom in a solid material is vibrating very rapidly about its lattice position within the crystal. In a sense, these **atomic vibrations** may be thought of as imperfections or defects. At any instant of time, not all atoms vibrate at the same frequency and amplitude or with the same energy. At a given temperature, there exists a distribution of energies for the constituent atoms about an average energy. Over time, the vibrational energy of any specific atom also varies in a random manner. With rising temperature, this average energy increases, and, in fact, the temperature of a solid is really just a measure of the average vibrational activity of atoms and molecules. At room temperature, a typical vibrational frequency is on the order of 10^{13} vibrations per second, whereas the amplitude is a few thousandths of a nanometer.

Many properties and processes in solids are manifestations of this vibrational atomic motion. For example, melting occurs when the vibrations are vigorous enough to rupture large numbers of atomic bonds. A more detailed discussion of atomic vibrations and their influence on the properties of materials is presented in Chapter 19.

Microscopic Examination

4.9 BASIC CONCEPTS OF MICROSCOPY

On occasion it is necessary or desirable to examine the structural elements and defects that influence the properties of materials. Some structural elements are of *macroscopic* dimensions; that is, they are large enough to be observed with the unaided eye. For example, the shape and average size or diameter of the grains for a polycrystalline specimen are important structural characteristics. Macroscopic grains are often evident on aluminum streetlight posts and also on highway guardrails. Relatively large grains having different textures are clearly visible on the surface of the sectioned copper ingot shown in Figure 4.13. However, in most materials the constituent grains are of *microscopic* dimensions, having diameters that may be on the order of microns,[10] and their details must be investigated using some type of microscope. Grain size and shape are only two features of what is termed the **microstructure;** these and other microstructural characteristics are discussed in subsequent chapters.

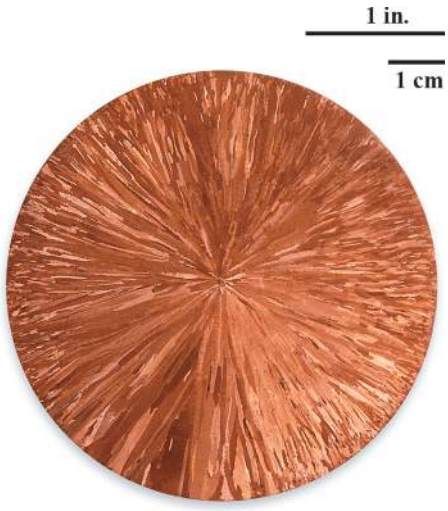
microstructure

Optical, electron, and scanning probe microscopes are commonly used in **microscopy.** These instruments aid in investigations of the microstructural features of all material types. Some of these techniques employ photographic equipment in conjunction with the microscope; the photograph on which the image is recorded is called a **photomicrograph.** In addition, many microstructural images are computer generated and/or enhanced.

microscopy

photomicrograph

Microscopic examination is an extremely useful tool in the study and characterization of materials. Several important applications of microstructural examinations are as follows: to ensure that the associations between the properties and structure (and defects) are properly understood, to predict the properties of materials once these relationships have been established, to design alloys with new property combinations, to determine whether a material has been correctly heat-treated, and to ascertain the mode of mechanical fracture. Several techniques that are commonly used in such investigations are discussed next.



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Figure 4.13 Cross-section of a cylindrical copper ingot. The small, needle-shape grains may be observed, which extend from the center radially outward.

[10]A micron (μm), sometimes called a *micrometer*, is 10^{-6} m.

4.10 MICROSCOPIC TECHNIQUES

Optical Microscopy

With *optical microscopy,* the light microscope is used to study the microstructure; optical and illumination systems are its basic elements. For materials that are opaque to visible light (all metals and many ceramics and polymers), only the surface is subject to observation, and the light microscope must be used in a reflecting mode. Contrasts in the image produced result from differences in reflectivity of the various regions of the microstructure. Investigations of this type are often termed *metallographic* because metals were first examined using this technique.

Normally, careful and meticulous surface preparations are necessary to reveal the important details of the microstructure. The specimen surface must first be ground and polished to a smooth and mirror-like finish. This is accomplished by using successively finer abrasive papers and powders. The microstructure is revealed by a surface treatment using an appropriate chemical reagent in a procedure termed *etching*. The chemical reactivity of the grains of some single-phase materials depends on crystallographic orientation. Consequently, in a polycrystalline specimen, etching characteristics vary from grain to grain. Figure 4.14*b* shows how normally incident light is reflected by three etched surface grains, each having a different orientation. Figure 4.14*a* depicts the surface structure as it might appear when viewed with the microscope; the luster or texture of each grain depends on its reflectance properties. A photomicrograph of a polycrystalline specimen exhibiting these characteristics is shown in Figure 4.14*c*.

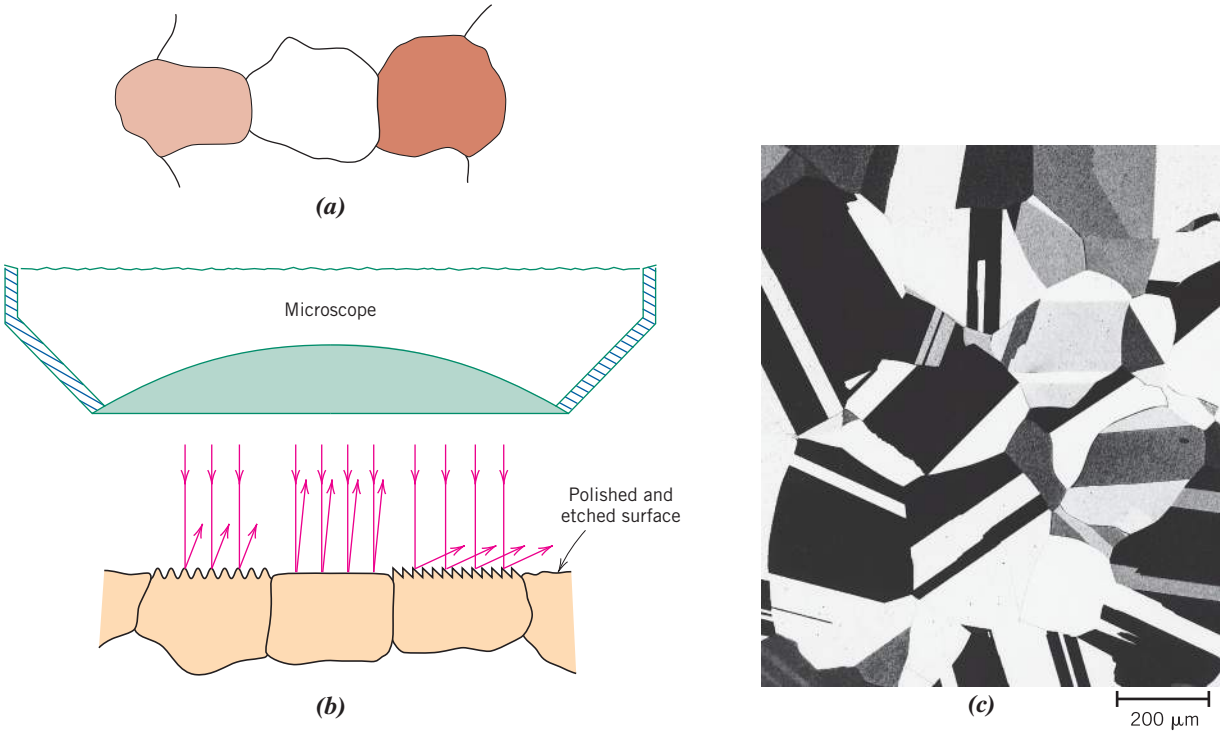


Figure 4.14 (*a*) Polished and etched grains as they might appear when viewed with an optical microscope. (*b*) Section taken through these grains showing how the etching characteristics and resulting surface texture vary from grain to grain because of differences in crystallographic orientation. (*c*) Photomicrograph of a polycrystalline brass specimen, 60×.
[Figure (*c*) from J. E. Burke, *Grain Control in Industrial Metallurgy*, in "The Fundamentals of Recrystallization and Grain Growth," Thirtieth National Metal Congress and Exposition, American Society for Metals, 1948. By permission of ASM International, Materials Park, OH. www.asmInternational.org.]

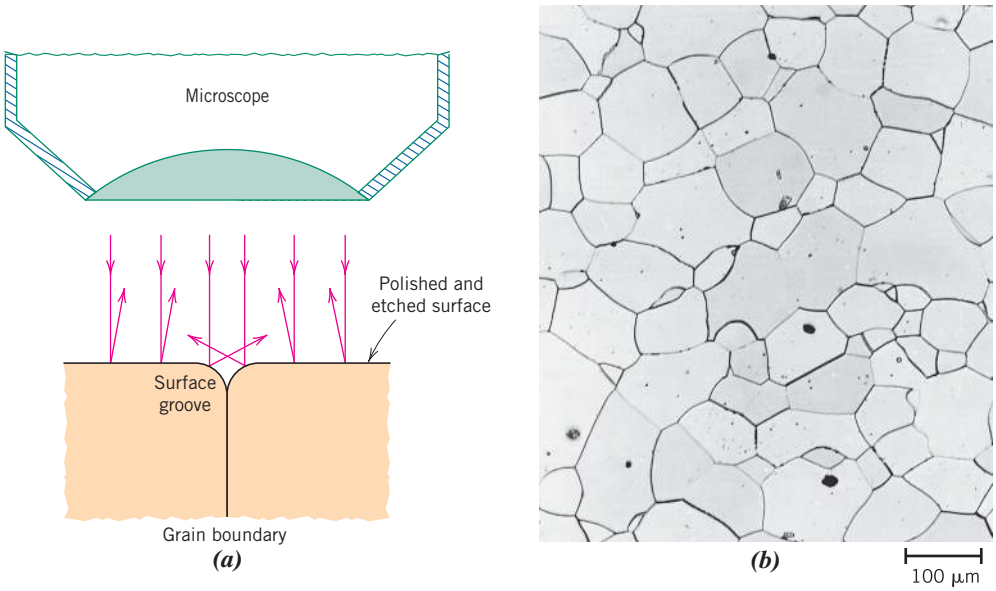


Figure 4.15 (*a*) Section of a grain boundary and its surface groove produced by etching; the light reflection characteristics in the vicinity of the groove are also shown. (*b*) Photomicrograph of the surface of a polished and etched polycrystalline specimen of an iron–chromium alloy in which the grain boundaries appear dark, 100×.
[Photomicrograph courtesy of L. C. Smith and C. Brady, the National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD.)]

Also, small grooves form along grain boundaries as a consequence of etching. Because atoms along grain boundary regions are more chemically active, they dissolve at a greater rate than those within the grains. These grooves become discernible when viewed under a microscope because they reflect light at an angle different from that of the grains themselves; this effect is displayed in Figure 4.15*a*. Figure 4.15*b* is a photomicrograph of a polycrystalline specimen in which the grain boundary grooves are clearly visible as dark lines.

When the microstructure of a two-phase alloy is to be examined, an etchant is often chosen that produces a different texture for each phase so that the different phases may be distinguished from each other.

Electron Microscopy

Some structural elements are too fine or small to permit observation using optical microscopy. The upper limit to the magnification possible with an optical microscope is approximately 2000×. Under such circumstances, the electron microscope, which is capable of much higher magnifications, may be employed.

An image of the structure under investigation is formed using beams of electrons instead of light radiation. According to quantum mechanics, a high-velocity electron becomes wavelike, having a wavelength that is inversely proportional to its velocity. When accelerated across large voltages, electrons can be made to have wavelengths on the order of 0.003 nm (3 pm). High magnifications and resolving powers of these microscopes are consequences of the short wavelengths of electron beams. The electron beam is focused and the image formed with magnetic lenses; otherwise, the geometry of the microscope components is essentially the same as with optical systems. Both transmission and reflection beam modes of operation are possible for electron microscopes.

Transmission Electron Microscopy

transmission electron microscope (TEM)

The image seen with a **transmission electron microscope (TEM)** is formed by an electron beam that passes through the specimen. Details of internal microstructural features are accessible to observation; contrasts in the image are produced by differences in beam scattering or diffraction produced between various elements of the microstructure or defect. Because solid materials are highly absorptive to electron beams, a specimen to be examined must be prepared in the form of a very thin foil; this ensures transmission through the specimen of an appreciable fraction of the incident beam. The transmitted beam is projected onto a fluorescent screen or a photographic film so that the image may be viewed. Magnifications approaching 1,000,000× are possible with transmission electron microscopy, which is frequently used to study dislocations.

Scanning Electron Microscopy

scanning electron microscope (SEM)

A more recent and extremely useful investigative tool is the **scanning electron microscope (SEM).** The surface of a specimen to be examined is scanned with an electron beam, and the reflected (or *back-scattered*) beam of electrons is collected and then displayed at the same scanning rate on a cathode ray tube (CRT; similar to a CRT television screen). The image on the screen, which may be photographed, represents the surface features of the specimen. The surface may or may not be polished and etched, but it must be electrically conductive; a very thin metallic surface coating must be applied to nonconductive materials. Magnifications ranging from 10× to in excess of 50,000× are possible, as are also very great depths of field. Accessory equipment permits qualitative and semiquantitative analysis of the elemental composition of very localized surface areas.

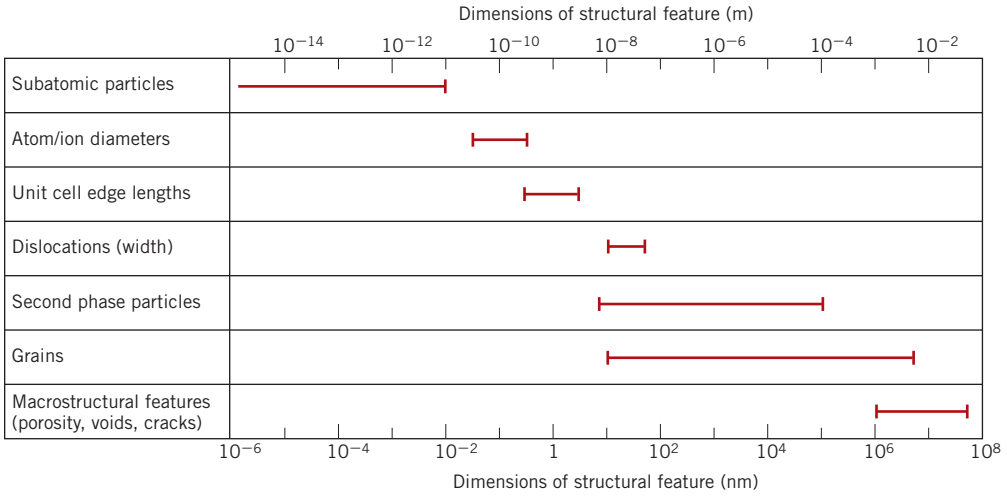
Scanning Probe Microscopy

scanning probe microscope (SPM)

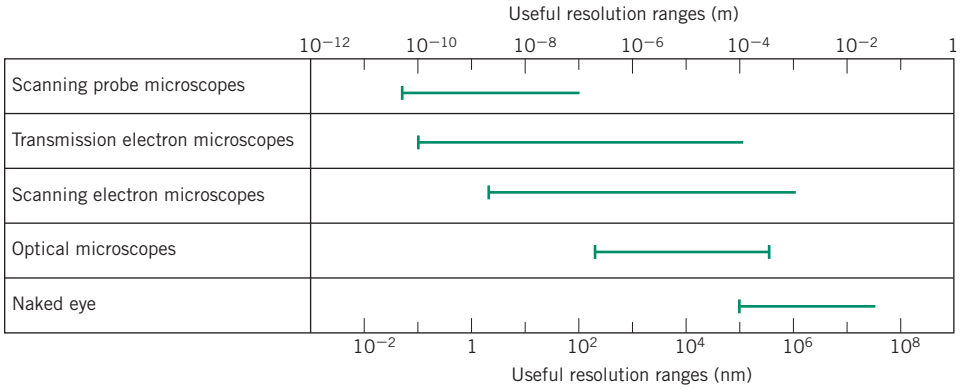
In the past two decades, the field of microscopy has experienced a revolution with the development of a new family of scanning probe microscopes. The **scanning probe microscope (SPM),** of which there are several varieties, differs from optical and electron microscopes in that neither light nor electrons are used to form an image. Rather, the microscope generates a topographical map, on an atomic scale, that is a representation of surface features and characteristics of the specimen being examined. Some of the features that differentiate the SPM from other microscopic techniques are as follows:

- Examination on the nanometer scale is possible inasmuch as magnifications as high as 10^9× are possible; much better resolutions are attainable than with other microscopic techniques.
- Three-dimensional magnified images are generated that provide topographical information about features of interest.
- Some SPMs may be operated in a variety of environments (e.g., vacuum, air, liquid); thus, a particular specimen may be examined in its most suitable environment.

Scanning probe microscopes employ a tiny probe with a very sharp tip that is brought into very close proximity (i.e., to within on the order of a nanometer) of the specimen surface. This probe is then raster-scanned across the plane of the surface. During scanning, the probe experiences deflections perpendicular to this plane in response to electronic or other interactions between the probe and specimen surface. The in-surface-plane and out-of-plane motions of the probe are controlled by piezoelectric (Section 18.25) ceramic components that have nanometer resolutions. Furthermore, these probe movements are monitored electronically and transferred to and stored in a computer, which then generates the three-dimensional surface image.



(a)



(b)

Figure 4.16 (*a*) Bar chart showing size ranges for several structural features found in materials. (*b*) Bar chart showing the useful resolution ranges for four microscopic techniques discussed in this chapter, in addition to the naked eye.
(Courtesy of Prof. Sidnei Paciornik, DCMM PUC-Rio, Rio de Janeiro, Brazil, and Prof. Carlos Pérez Bergmann, Federal University of Rio Grande do Sul, Porto Alegre, Brazil.)

These new SPMs, which allow examination of the surface of materials at the atomic and molecular level, have provided a wealth of information about a host of materials, from integrated circuit chips to biological molecules. Indeed, the advent of the SPMs has helped to usher in the era of *nanomaterials*—materials whose properties are designed by engineering atomic and molecular structures.

Figure 4.16*a* is a bar chart showing dimensional size ranges for several types of structures found in materials (note that the axes are scaled logarithmically). The useful dimensional resolution ranges for the several microscopic techniques discussed in this chapter (plus the naked eye) are presented in the bar chart of Figure 4.16*b*. For three of these techniques (SPM, TEM, and SEM), an upper resolution value is not imposed by the characteristics of the microscope and, therefore, is somewhat arbitrary and not well defined. Furthermore, by comparing Figures 4.16*a* and 4.16*b*, it is possible

to decide which microscopic technique(s) is (are) best suited for examination of each of the structure types.

4.11 GRAIN-SIZE DETERMINATION

grain size

The **grain size** is often determined when the properties of polycrystalline and single-phase materials are under consideration. In this regard, it is important to realize that for each material, the constituent grains have a variety of shapes and a distribution of sizes. Grain size may be specified in terms of average or mean grain diameter, and a number of techniques have been developed to measure this parameter.

Before the advent of the digital age, grain-size determinations were performed manually using photomicrographs. However, today, most techniques are automated and use digital images and image analyzers with the capacity to record, detect, and measure accurately features of the grain structure (i.e., grain intercept counts, grain boundary lengths, and grain areas).

We now briefly describe two common grain-size determination techniques: (1) *linear intercept*—counting numbers of grain boundary intersections by straight test lines; and (2) *comparison*—comparing grain structures with standardized charts, which are based upon grain areas (i.e., number of grains per unit area). Discussions of these techniques is from the manual perspective (using photomicrographs).

For the linear intercept method, lines are drawn randomly through several photomicrographs that show the grain structure (all taken at the same magnification). Grain boundaries intersected by all the line segments are counted. Let us represent the sum of the total number of intersections as P and the total length of all the lines by L_T. The mean intercept length $\bar{\ell}$ [in real space (at 1×—i.e., not magnified)], a measure of grain diameter, may be determined by the following expression:

$$\bar{\ell} = \frac{L_T}{PM} \tag{4.16}$$

where M is the magnification.

The comparison method of grain-size determination was devised by the American Society for Testing and Materials (ASTM).[11] The ASTM has prepared several standard comparison charts, all having different average grain sizes and referenced to photomicrographs taken at a magnification of 100×. To each chart is assigned a number ranging from 1 to 10, which is termed the *grain-size number*. A specimen must be prepared properly to reveal the grain structure, which is then photographed. Grain size is expressed as the grain-size number of the chart that most nearly matches the grains in the micrograph. Thus, a relatively simple and convenient visual determination of grain-size number is possible. Grain-size number is used extensively in the specification of steels.

Relationship between ASTM grain size number and number of grains per square inch (at 100×)

The rationale behind the assignment of the grain-size number to these various charts is as follows: Let G represent the grain-size number, and let n be the average number of grains per square inch at a magnification of 100×. These two parameters are related to each other through the expression[12]

$$n = 2^{G-1} \tag{4.17}$$

[11]ASTM Standard E112, "Standard Test Methods for Determining Average Grain Size."

[12]Please note that in this edition, the symbol n replaces N from previous editions; also, G in Equation 4.17 is used in place of the previous n. Equation 4.17 is the standard notation currently used in the literature.

For photomicrographs taken at magnifications other than 100×, use of the following modified form of Equation 4.17 is necessary:

$$n_M\left(\frac{M}{100}\right)^2 = 2^{G-1} \tag{4.18}$$

In this expression, n_M is the number of grains per square inch at magnification M. In addition, the inclusion of the $\left(\frac{M}{100}\right)^2$ term makes use of the fact that, whereas magnification is a length parameter, area is expressed in terms of units of length squared. As a consequence, the number of grains per unit area increases with the square of the increase in magnification.

Relationships have been developed that relate mean intercept length to ASTM grain-size number; these are as follows:

$$G = -6.6457 \log \bar{\ell} - 3.298 \qquad \text{(for } \bar{\ell} \text{ in mm)} \tag{4.19a}$$

$$G = -6.6353 \log \bar{\ell} - 12.6 \qquad \text{(for } \bar{\ell} \text{ in in.)} \tag{4.19b}$$

At this point, it is worthwhile to discuss the representation of magnification (i.e., linear magnification) for a micrograph. Sometimes magnification is specified in the micrograph legend (e.g., "60×" for Figure 4.14*b*); this means the micrograph represents a 60 times enlargement of the specimen in real space. *Scale bars* are also used to express degree of magnification. A scale bar is a straight line (typically horizontal), either superimposed on or located near the micrograph image. Associated with the bar is a length, typically expressed in microns; this value represents the distance in magnified space corresponding to the scale line length. For example, in Figure 4.15*b*, a scale bar is located below the bottom right-hand corner of the micrograph; its "100 μm" notation indicates that 100 μm correlates with the scale bar length.

To compute magnification from a scale bar, the following procedure may be used:

1. Measure the length of the scale bar in millimeters using a ruler.
2. Convert this length into microns [i.e., multiply the value in step (1) by 1000 because there are 1000 microns in a millimeter].
3. Magnification M is equal to

$$M = \frac{\text{measured scale length (converted to microns)}}{\text{the number appearing by the scale bar (in microns)}} \tag{4.20}$$

For example, for Figure 4.15*b*, the measured scale length is approximately 10 mm, which is equivalent to (10 mm)(1000 μm/mm) = 10,000 μm. Inasmuch as the scale bar length is 100 μm, the magnification is equal to

$$M = \frac{10{,}000\ \mu\text{m}}{100\ \mu\text{m}} = 100\times$$

This is the value given in the figure legend.



Concept Check 4.4 Does the grain-size number (G of Equation 4.17) increase or decrease with decreasing grain size? Why?

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]